

Databricks Lab : Delta Lake OPTIMIZE Command

Duration : 60 Min.

Scenario

You need to perform performance optimization on a Delta table containing fragmented files using the OPTIMIZE command.

Input Details

The environment has been pre-configured with the following resources: - **Workspace:** Pre-configured through KodeBuck Service credentials (DATABRICKS_HOST, DATABRICKS_TOKEN)

Username & Naming Conventions

Your candidate prefix is a combination of your username (the part of your email/login before the @ or _ symbol) and exam code. For example, if your username is student (or student_labskraft.com) and your exam code is exam123, your prefix is **student-exam123**.

You must name your resources accordingly:

- Delta Table:** <prefix>_optimize_table (e.g. student_exam123_optimize_table)

Task Objectives

1. Table Optimization

- Create a Delta table named <prefix>_optimize_table.
- Execute the OPTIMIZE statement to compact small files.

Verification

Once you have performed the tasks, you can run the verification script to check your progress and receive your score. The verification system will check the actual resources in the environment.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Delta table compact files check	4 Marks
TC2	Query history compaction log verify	4 Marks

TC3	Schema integrity validation	4 Marks
TC4	Reserved validation	4 Marks
TC5	Reserved validation	4 Marks

Total Score: 20 Marks

Important Notes

- Ensure all resource names contain your candidate username prefix in lowercase.
- Check that your script compiles cleanly and contains no syntax errors.